

Justify your answers and show your work.

Evaluate each algebraic expression for the given value of the variable.

1. $4 - 2(3 + x)^2$, for $x = -5$

2. $\frac{2x+y}{xy-2x}$, for $x = 1$ and $y = -4$

Simplify the algebraic expression. Write all exponents as positive.

3. $7 - 6[-\frac{1}{2}(6 - 2) + 1]$

4. $\frac{5}{(7-12)^2} + \frac{1+3}{5}$

5. $-\left(\frac{4}{y}\right)^2$

6. $(4x^2)^{1/2}$

7. $(x^3x^2)^{-2}$

8. $(-2x^3y)(-6x^6y^{10})$

9. $\frac{6xy^3}{18x^2y^{-5}}$

10. $\sqrt{\frac{1}{25}}$

11. $5\sqrt{12} - 3\sqrt{3}$

Perform the indicated operation and simplify.

12. $(x^5 - 8x^2 + 5x) - (-2x^5 + x^2 + 12)$

13. $(2x^2y - 2xy) + (3x^2y - xy)$

14. $(6x - 2)(x^2 + x - 3)$

15. $(5x + 3y)^2$

Factor Completely.

16. $x^2 + 3x - 10$

17. $3x^3 - 6x^2$

18. $3x^3 + 9x^2 - 30x$

19. $16y^2 - 25$

Factor by grouping.

20. $x^3 - 3x^2 + 4x - 12.$